



HA NOI UNIVERSITY OF SCIENCE AND TECHNOLOGY
SCHOOL OF INFORMATION AND COMMUNICATION TECHNOLOGY

Computer Vision

Chapter 5: Image segmentation

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Chapter 5. Segmentation

- Introduction to image segmentation
- Segmentation based on pixel classification
 - Thresholding
 - Clustering techniques
- Region-based segmentation
 - Region growing algorithm,
 - Split and merge algorithm
- Edge-based segmentation

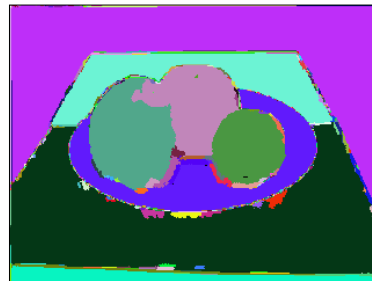


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Introduction

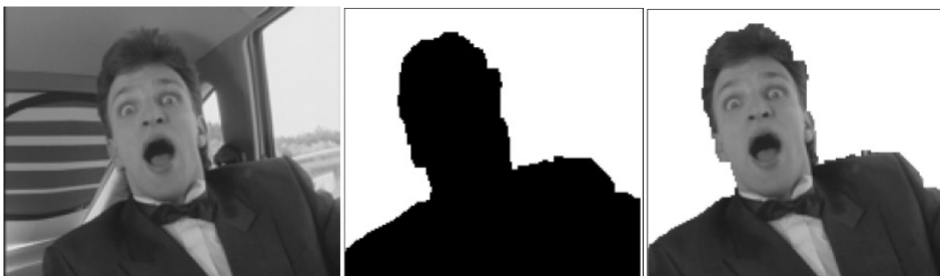
- Purpose:
 - to partition an image into meaningful regions with respect to a particular application
- Goal:
 - to cluster pixels into salient image regions, i.e., regions corresponding to individual surfaces, objects, or natural parts of objects.
- The segmentation is based on the feature measurements taken from the image:
 - grey level, color, texture, depth or motion...

Introduction



Introduction

- Entity can be extracted from images using mask



Source : Pascal Bertolino, Cours de Traitement d'images. LIS, INPG (France)

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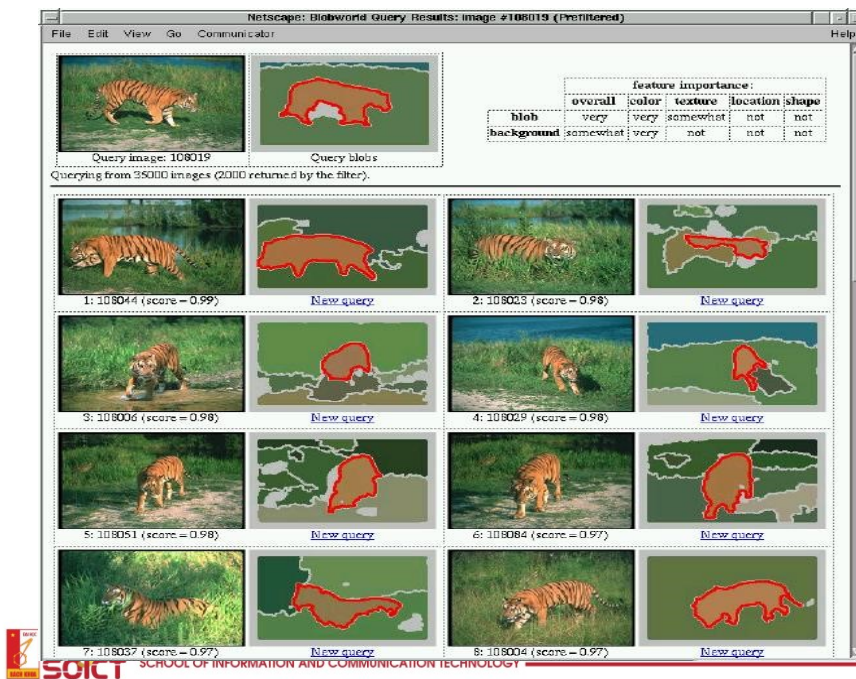
Applications

- **Image segmentation**
 - is usually an initial and vital step in a series of processes aimed at overall image understanding of computer vision
- **Segmentation applications:**
 - Object recognition;
 - Image retrieval;
 - Medical image analysis;
 - Boundary estimation within motion or stereo systems;
 - Tracking of objects in a video;
 - Classification of terrains visible in satellite images
 - ...



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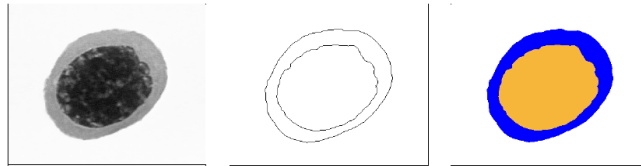
Approaches for image segmentation

- Segmentation is usually based on:
 - **discontinuities**: edges
 - sudden changes, borders (frontiers) between regions...
 - **homogeneous** zones: regions
 - same color, texture, intensity, ...

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Approaches for image segmentation

- Pixel-based approach
- Region-based approach:
 - look for **homogeneous** areas in the image
- Edge-based approach :
 - look for **discontinuities** in the image
 - **A closed edge is equivalent to a region**
- Hybrid (Dual) approach (region + edge)

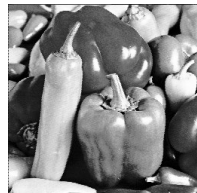
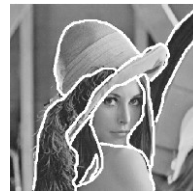


Examples

Original images



Segmented images



Pixel-based approach

- Pixel-based approach
 - Thresholding
 - Clustering
- It is not a region segmentation technique
 - But we often in segmentation looking for regions
 - Need some post-processing

Thresholding

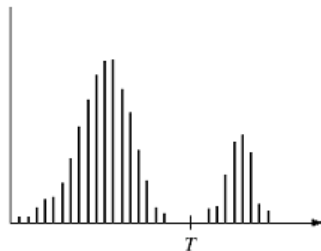
- Thresholding is a *simple and popular* method for object segmentation in digital images
- Thresholding can be
 - *Global*: one threshold for the whole image
 - *Local*: one threshold for a part of the image
 - *Adaptive*: one threshold adjusted according to each image or each image part

Basic global thresholding

- Basic thresholding (2 classes) – main idea :
 - IF value(pixel) $\geq$ threshold THEN value(pixel) = 1 (or 255)
 - IF value(pixel) < threshold THEN value(pixel) = 0
- The result is a binary image
- It is also possible to use n thresholds to split the image in $n+1$ classes
- Problem: choosing the threshold(s)!

Basic global thresholding

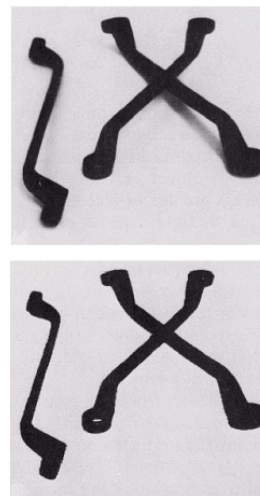
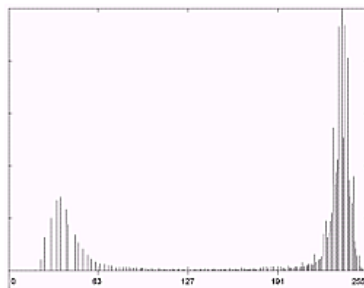
- Find the threshold on histogram of gray level intensity (histogram thresholding)



$$g(x, y) = \begin{cases} 0, & f(x, y) < T \\ 1, & f(x, y) \geq T \end{cases}$$

Basic global thresholding

- Threshold value: not difficult if
 - Controlled environment
 - Industrial applications

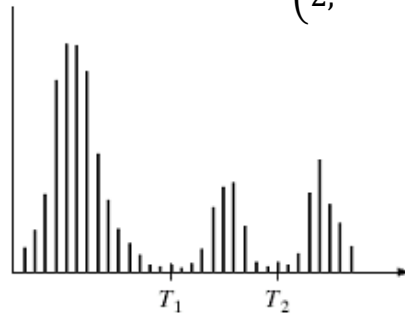


Multi-thresholds

- n thresholds to split the image in $n+1$ classes:
 - IF $\text{value}(\text{pixel}) < \text{threshold_1}$
 - THEN $\text{value}(\text{pixel}) = 0$
 - IF $\text{value}(\text{pixel}) \geq \text{threshold_1} \ \&\& \ \text{value}(\text{pixel}) < \text{threshold_2}$
 - THEN $\text{value}(\text{pixel}) = 1$
 - ...
 - IF $\text{value}(\text{pixel}) \geq \text{threshold_n}$
 - THEN $\text{value}(\text{pixel}) = n$
- Problems: **How many thresholds?**

Multi-thresholds

$$g(x, y) = \begin{cases} 0, & f(x, y) < T_1 \\ 1, & T_1 \leq f(x, y) < T_2 \\ 2, & f(x, y) \geq T_2 \end{cases}$$

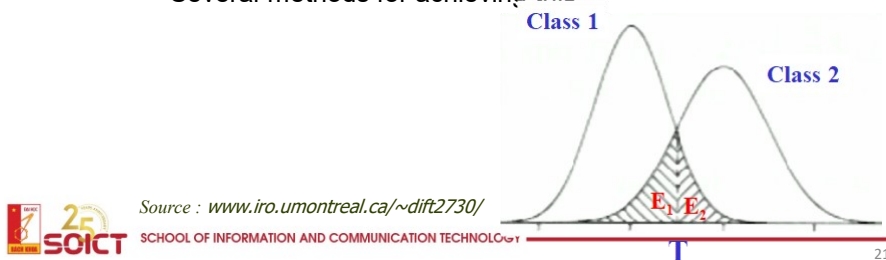


Threshold value

- Global thresholding: How to find the value of the threshold **T** ?
 - Value obtained **by tests**
 - The **mean** value of gray values
 - The **median** value between the min gray level and the max one
 - **One value balancing** both sections of the histogram
 - [automatic thresholding](#)

Choice of thresholds (optimal)

- 2 surfaces (background and object) in an image
 - We suppose mathematical models for distributions (gaussians, etc.)
 - We determine the probability of error in the classes 1 and 2 (surfaces 1 et 2)
 - We search for a threshold T resulting in a minimum error
 - Several methods for achieving this



Example: Global automatic thresholding

- One possible algorithm:
 - Choose an initial value for the threshold T (mean, median, ...)
 - We obtain 2 groups of pixels
 - $G1$ where $f(x,y) \geq T$ and $G2$ where $f(x,y) < T$
 - Compute the gray level means for $G1$ and $G2 \rightarrow \mu_1$ and μ_2
 - Compute a new value for T
 - $T = 1/2 (\mu_1 + \mu_2)$
 - Repeat until T is ~ constant
- There exist many other global automatic methods
 - Otsu, Kittler, K-means, ...
 - No solution on which one to use
 - Must be tested for each new application

Example: Otsu algorithm

- Sweep all possible threshold value for **T**
- For **each value of T**:

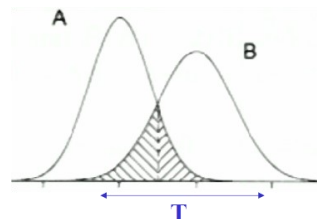
- Compute the mean and variance for each class
- We look for the intraclass variance

- Means: μ_1, μ_2
- Variances: σ_1^2, σ_2^2
- Intra-class variance:

$$\sigma_w^2 = P_1 * \sigma_1^2 + P_2 * \sigma_2^2$$

- The optimal threshold is the one with the **minimum value for σ_w^2**

- It is based on the idea that classes are well defined and well grouped



$$\sigma_1^2 = \frac{1}{T} \sum_{i=0}^{T-1} (h(i) - \mu_1)^2$$

$$\sigma_2^2 = \frac{1}{256 - T} \sum_{i=T}^{255} (h(i) - \mu_2)^2$$

$$\mu_1 = \frac{1}{T} \sum_{i=0}^{T-1} h(i) \quad P_1 = \frac{1}{N \cdot M} \sum_{i=0}^{T-1} h(i)$$

$$\mu_2 = \frac{1}{256 - T} \sum_{i=T}^{255} h(i) \quad P_2 = \frac{1}{N \cdot M} \sum_{i=T}^{255} h(i)$$



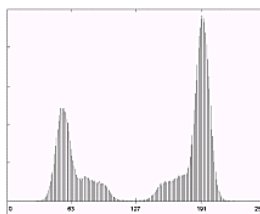
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Source : www.iro.umontreal.ca/~dift2730/

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Example: Otsu algorithm

- Threshold found by the algorithm:
 - T = 125

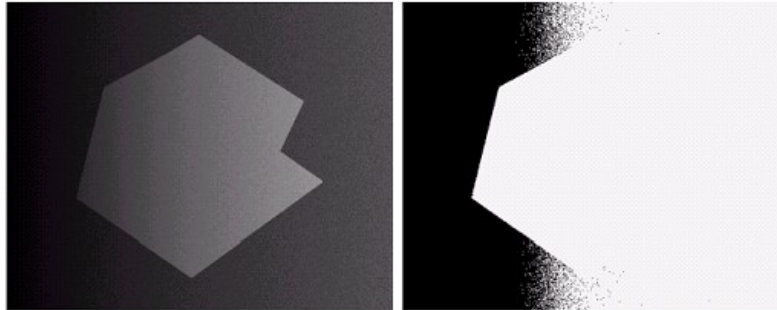


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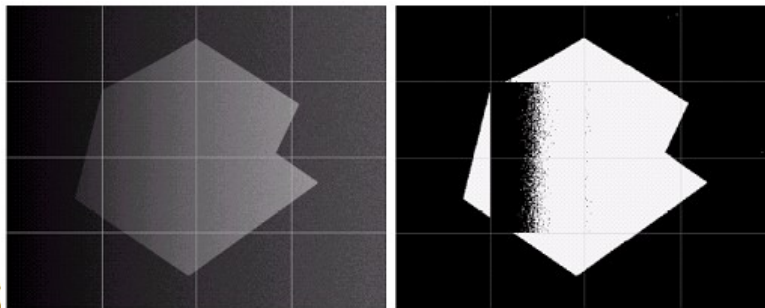
Global threshold: problem

- Problem:
 - Global thresholding cannot be used in that case
 - Solution: adaptive local thresholding

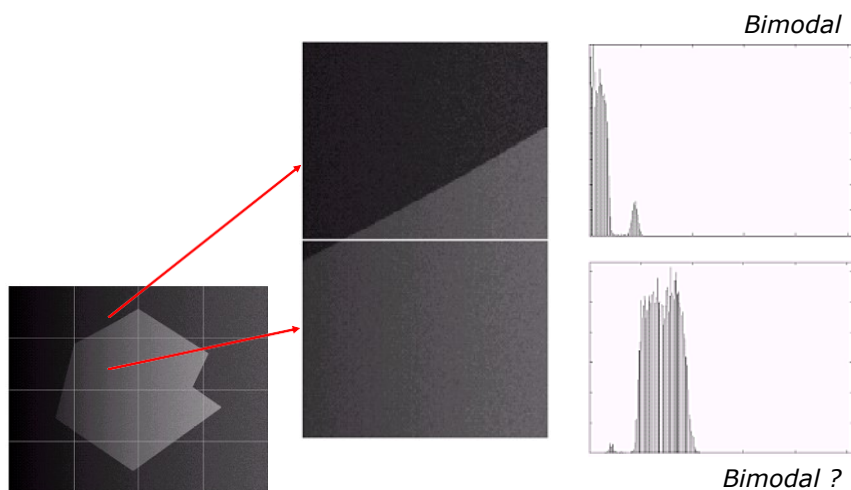


Example of adaptive thresholding

- Split the image in sub-images and process each sub-image with its own threshold
- The main decision is to choose the size of the sub-images
- Before processing each sub-image, check the variance to make sure that the sub-image contains two regions, and not only one.
 - Example: no thresholding for a sub-image if variance < 100



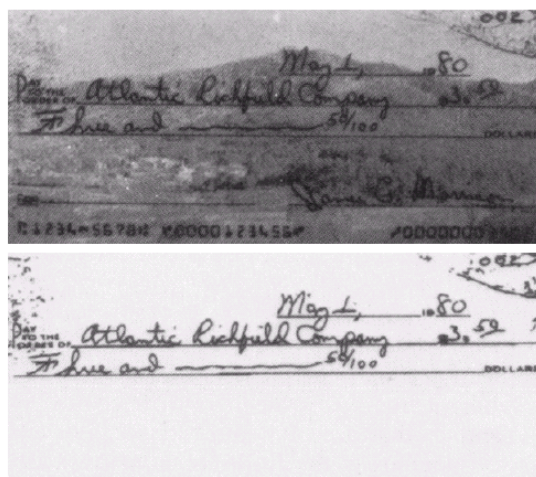
Example of adaptive thresholding



Example of adaptive thresholding

a
b

FIGURE 10.37
(a) Original image. (b) Image segmented by local thresholding. (Courtesy of IBM Corporation.)



Clustering-based segmentation

- Image is considered as a set of N image pixels.
- Attributes (property) of the pixels
 - gray level of single-band gray tone images,
 - color values of three-band color images: (r, g, b)
 - values of multi-band images, ...
- Based on the similar attribute, pixels classification operators partition an image into homogeneous regions.
 - Clustering provides a grouping of the pixels that is dependent on their values in the image but not necessarily on their locations in the image unless location is a specified property
 - Classifier provide the pixel classes which should be homogeneous regions.



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Clustering algorithms

- **Image segmentation approaches including:**
 - Feature space clustering approaches
 - Graph-based approaches
- **Clustering algorithms:**
 - K-Means clustering
 - Mean-Shift Clustering
 - Expectation-Maximization Clustering
 - Watershed Segmentation
 - Graph Cuts (Spectral clustering)
 - Normalized cuts: Jianbo Shi and J. Malik, "Normalized cuts and image segmentation, TPAMI2000
 - ...

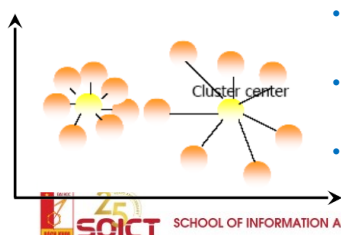


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K-means Clustering

- Let $X = \{p_1, \dots, p_N\}$ be a set of N image pixels:
 - $V(p_i)$: the property vector associated with pixel p_i
 - The clustering algorithm is to partition the image into K clusters (K regions)
- The **K-means** algorithm:
 - Initialization step: An initial property vector of each class C_k is chosen randomly from the set of all property vector, note $\mu_k(0)$
 - Interactive step: Assignment of image pixels to K clusters

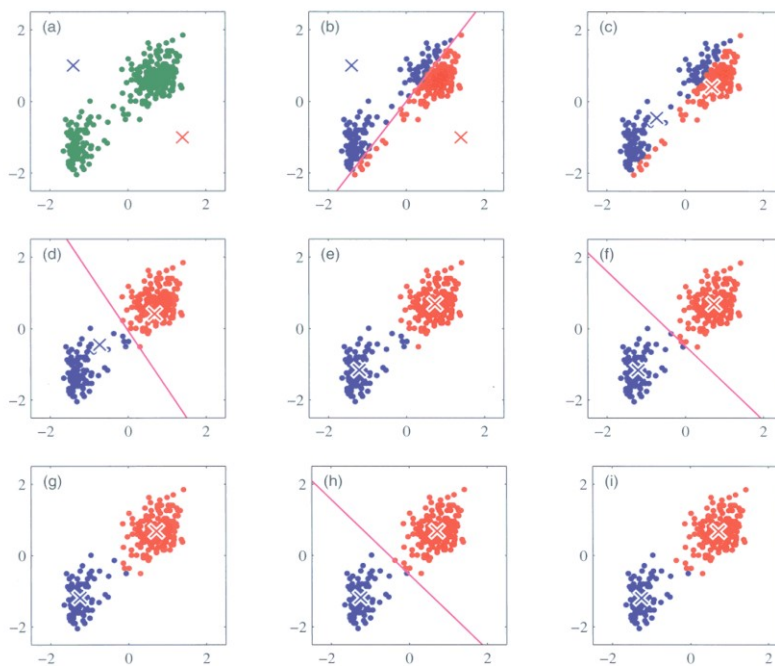


- Pixel p_i is assigned to the closest cluster, using a distance between 2 property vectors.
- Update the property vector of each class: $\mu_k(t)$ is computed as the mean of $\{V(p_i) \mid x_i \in C_k\}$.
- 2 Steps above are repeated until algorithm convergence.
- each class C_k should be a region R_k



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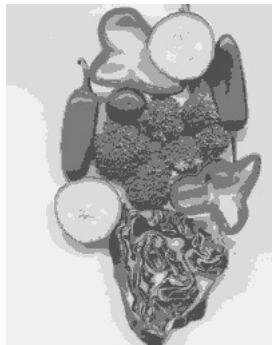


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K-means Clustering



Input image



K-means on gray level



K-means on color



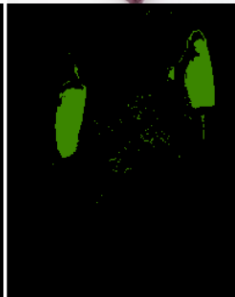
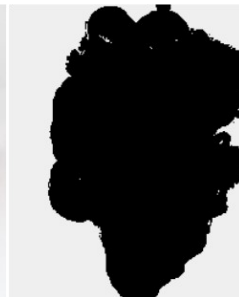
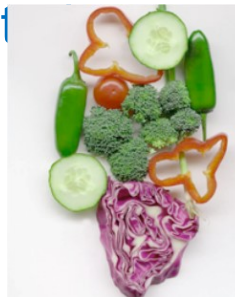
Source : D.A. Forsyth and J. Ponce. *Computer Vision : A Modern Approach*. Prentice-Hall, 2002.

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K-means Clust

K-means on color for
11 groups

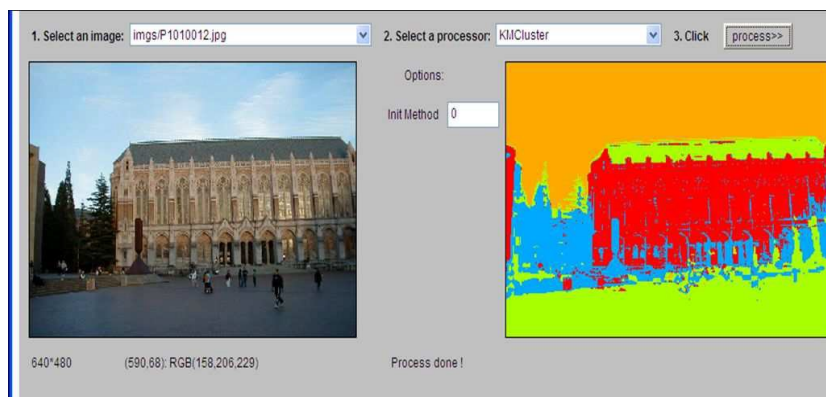


Source : D.A. Forsyth and J. Ponce. *Computer Vision : A Modern Approach*. Prentice-Hall, 2002.

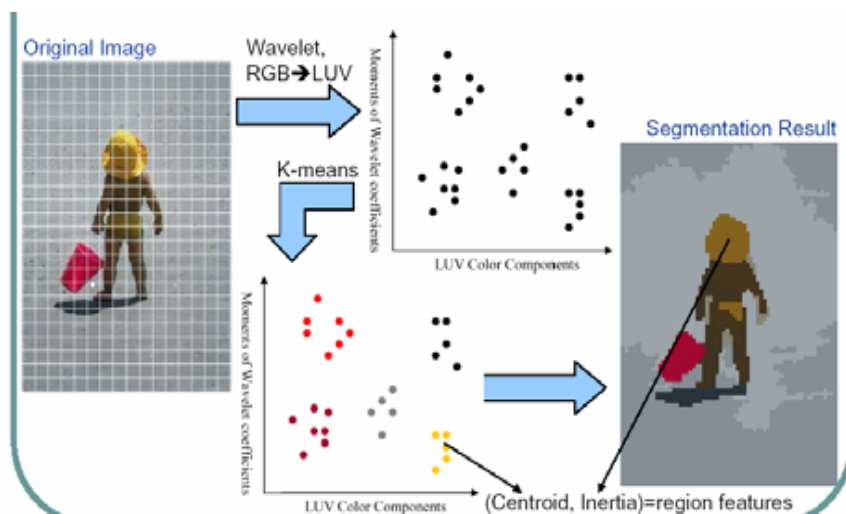
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Example



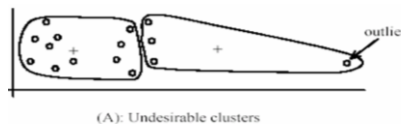
Example



K-means pros and cons

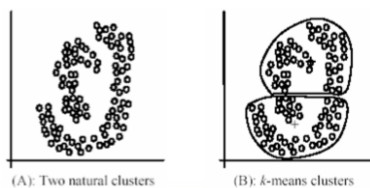
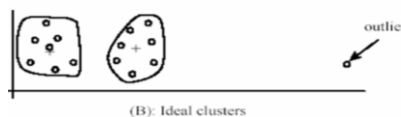
Pros

- Simple
- Converges to local minimum of within-cluster squared error



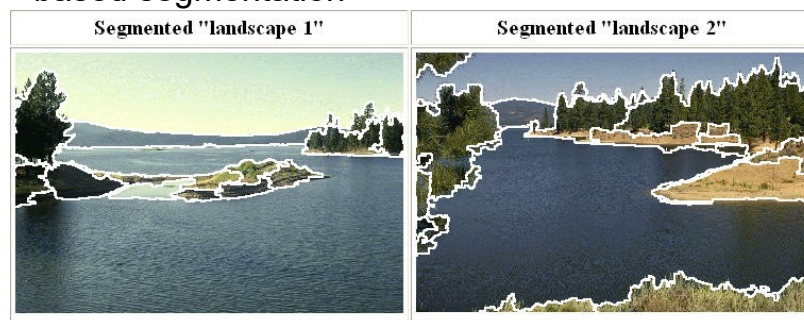
Cons/issues

- **Setting k?**
- Sensitive to initial centers
- Sensitive to outliers
- **Detects spherical clusters**
- Assuming means can be computed



Mean shift clustering and segmentation

- An advanced and versatile technique for clustering-based segmentation



<http://www.caip.rutgers.edu/~comanici/MSPAMI/msPamiResults.html>

D. Comaniciu and P. Meer, [Mean Shift: A Robust Approach toward Feature Space Analysis](#), PAMI 2002.

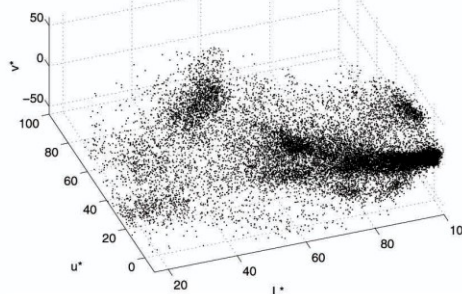
Mean shift algorithm

- The mean shift algorithm seeks *modes* or local maxima of density in the feature space

image

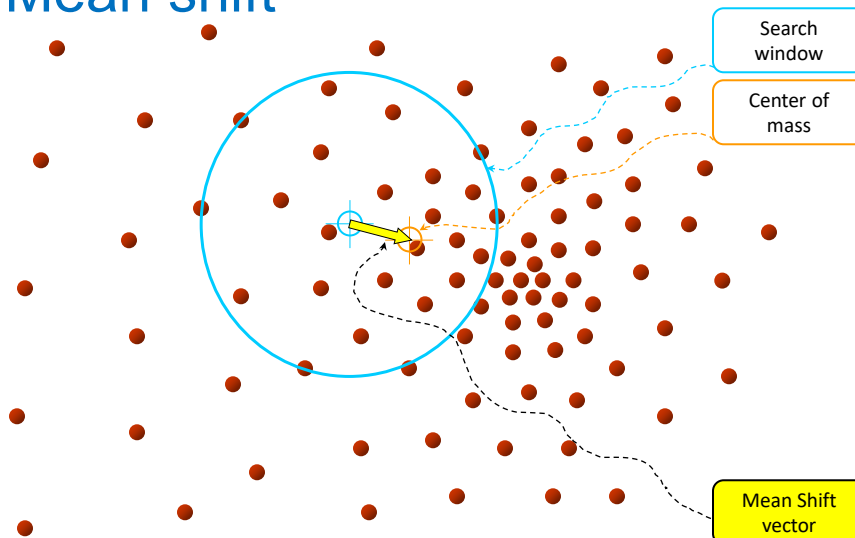


Feature space
($L^*u^*v^*$ color values)



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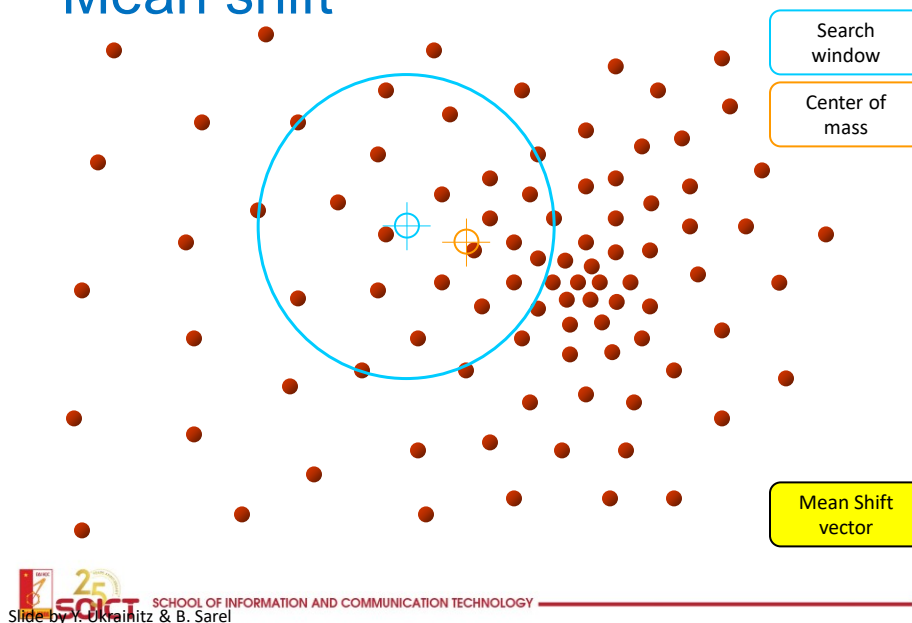
Mean shift



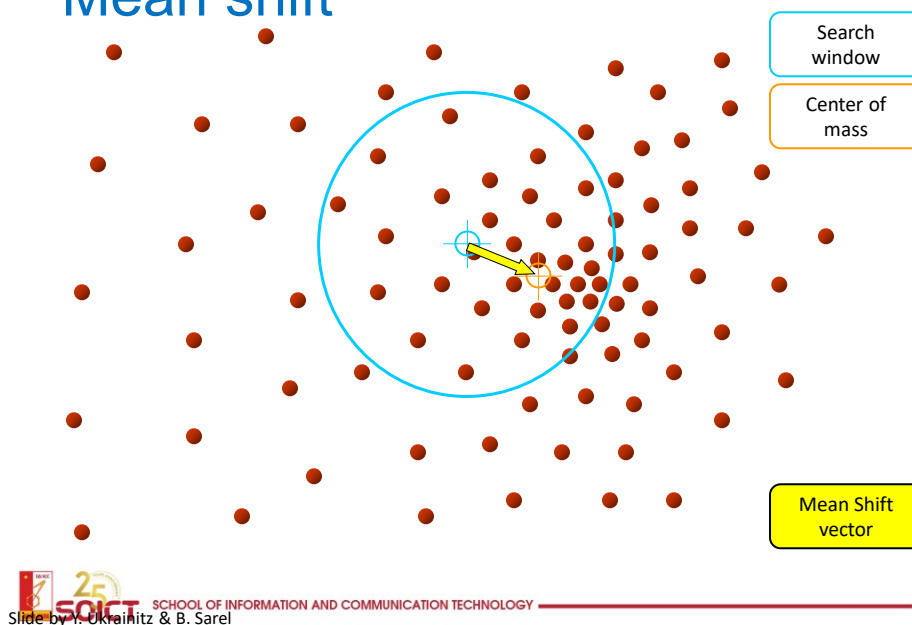
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Slide by V. Ukrainitz & B. Sarel

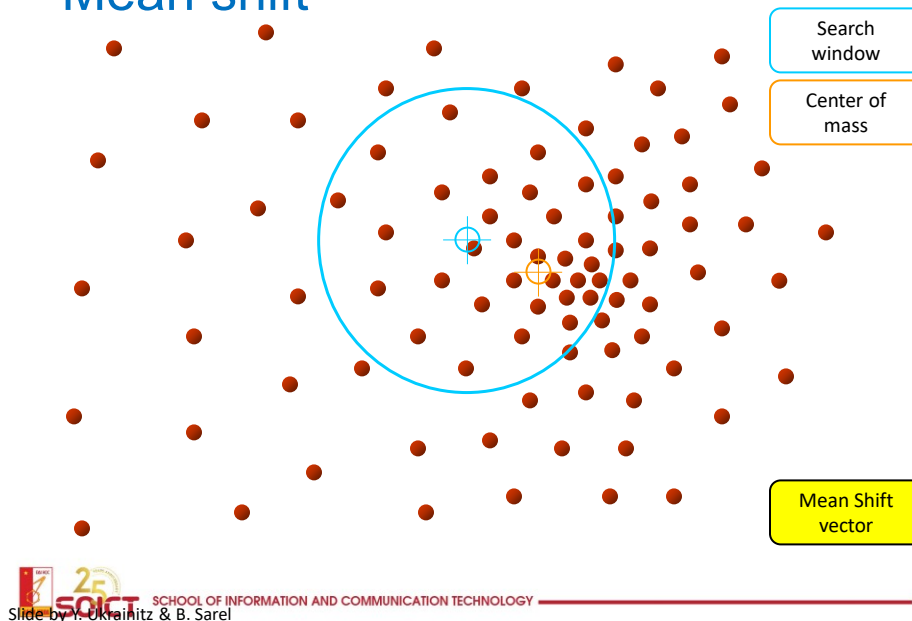
Mean shift



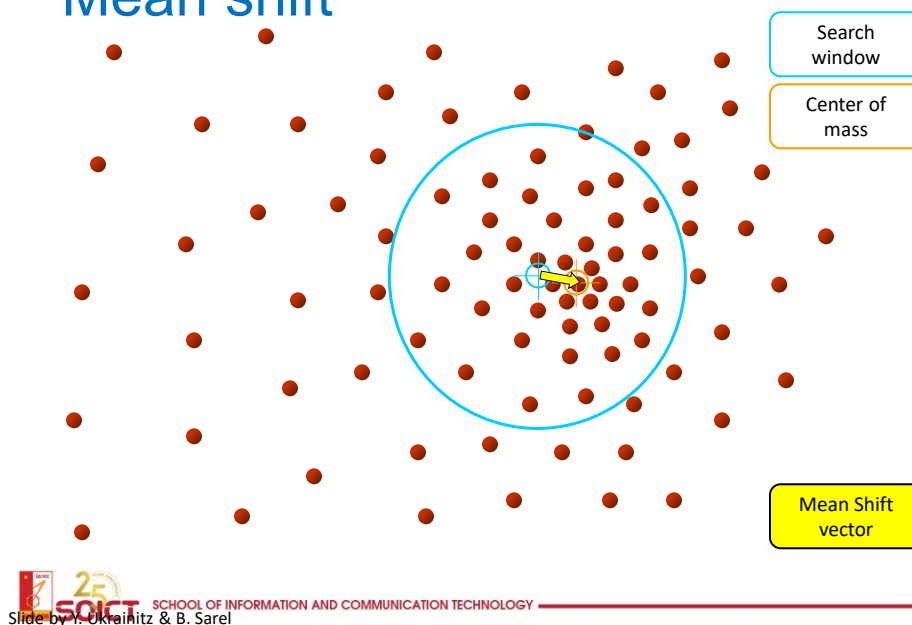
Mean shift



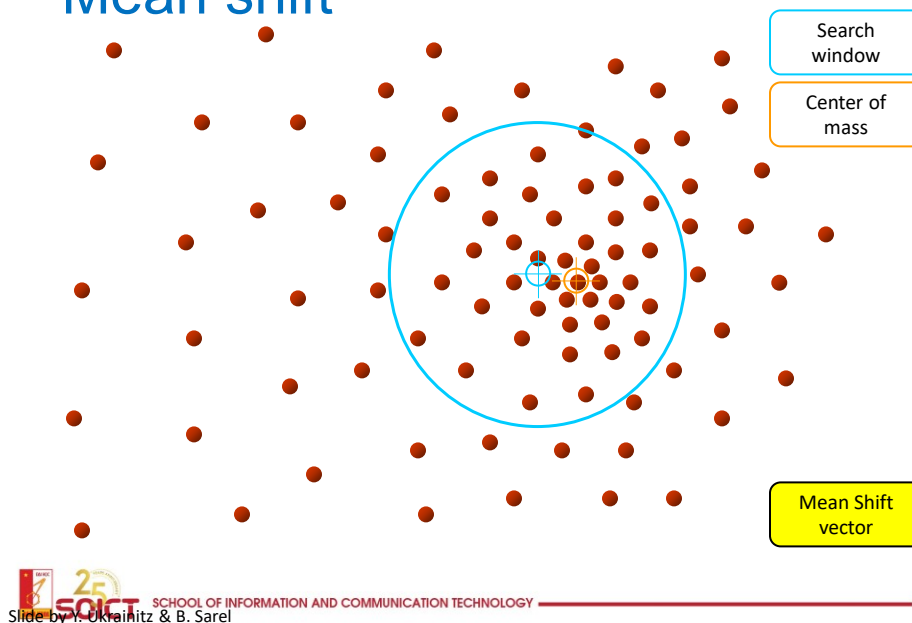
Mean shift



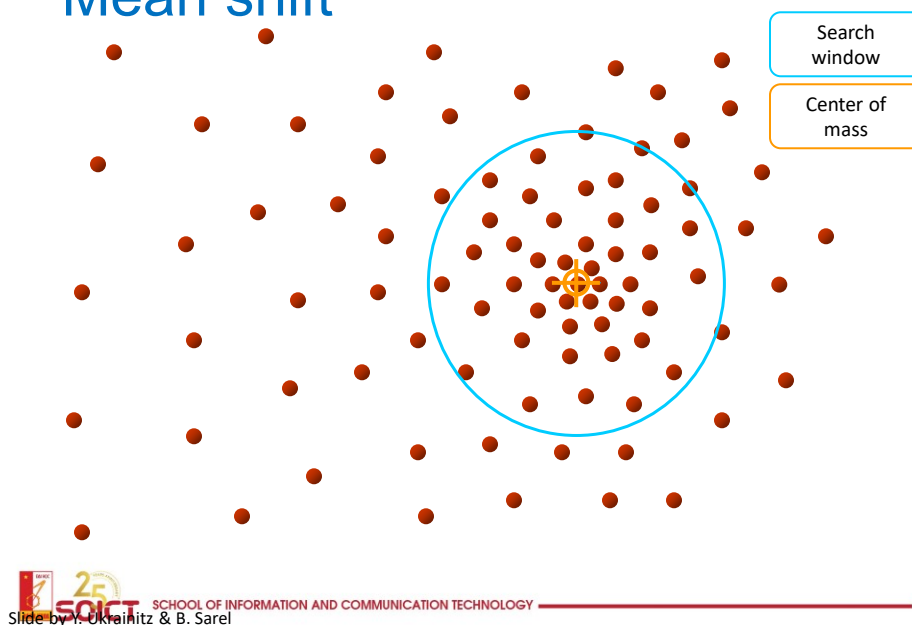
Mean shift



Mean shift

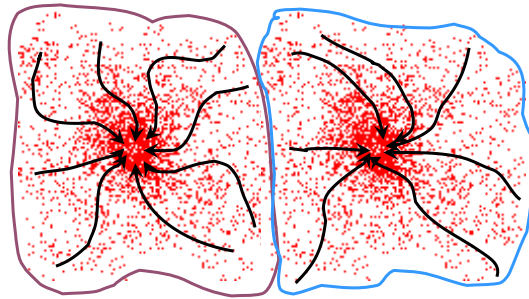


Mean shift



Mean shift clustering

- Cluster: all data points in the attraction basin of a mode
- Attraction basin: the region for which all trajectories lead to the same mode

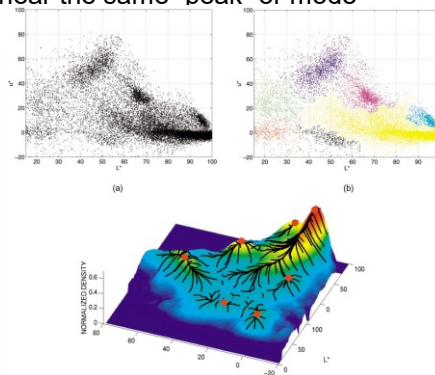


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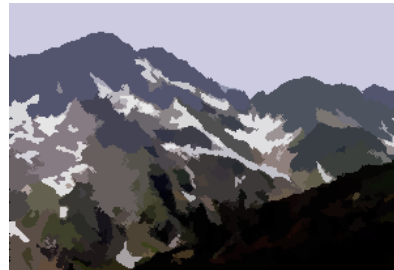
Mean shift clustering/segmentation

- Find features (color, gradients, texture, etc)
- Initialize windows at individual feature points
- Perform mean shift for each window until convergence
- Merge windows that end up near the same “peak” or mode



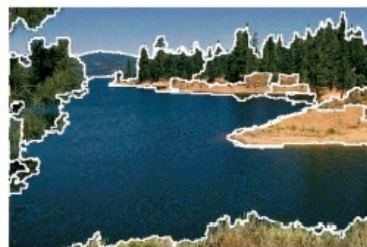
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Mean shift segmentation results



<http://www.caip.rutgers.edu/~romanici/MSPAMI/msPamiResults.html>

Mean shift segmentation results



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Mean shift

- Pros:
 - Does not assume shape on clusters
 - One parameter choice (window size, aka “bandwidth”)
 - Generic technique
 - Find multiple modes
- Cons:
 - Selection of window size
 - Does not scale well with dimension of feature space



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Pixel-based approach: Pros & cons

- Pros
 - Simple
- Cons: thresholding is mainly an operation on pixels
 - It does **not give connected regions** → can add more features
 - we need to « clean » the results
 - erase **lonely pixels**, keep regions
- Other segmentation methods exist
 - that can keep the integrity of regions (connected pixels)



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Features for segmentation

- Intensity, Color?
- Position
- Texture
- ...



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Segmentation as clustering

Depending on what we choose as the *feature space*, we can group pixels in different ways.

Grouping pixels based on **intensity** similarity



Feature space: intensity value (1-d)



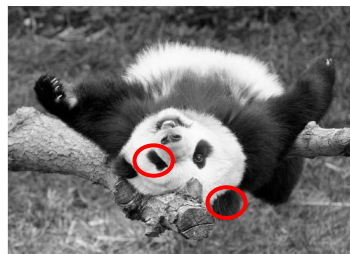
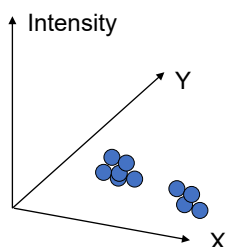
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Slide credit: Kristen Grauman

Segmentation as clustering

Depending on what we choose as the *feature space*, we can group pixels in different ways.

Grouping pixels based on **intensity+position** similarity



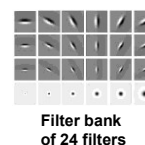
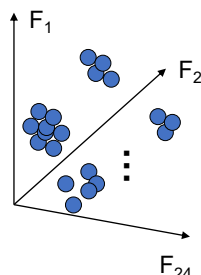
Both regions are black, but if we also include **position (x,y)**, then we could group the two into distinct segments; way to encode both **similarity & proximity**.

Slide credit: Kristen Grauman
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Segmentation as clustering

Depending on what we choose as the *feature space*, we can group pixels in different ways.

Grouping pixels based on **texture** similarity



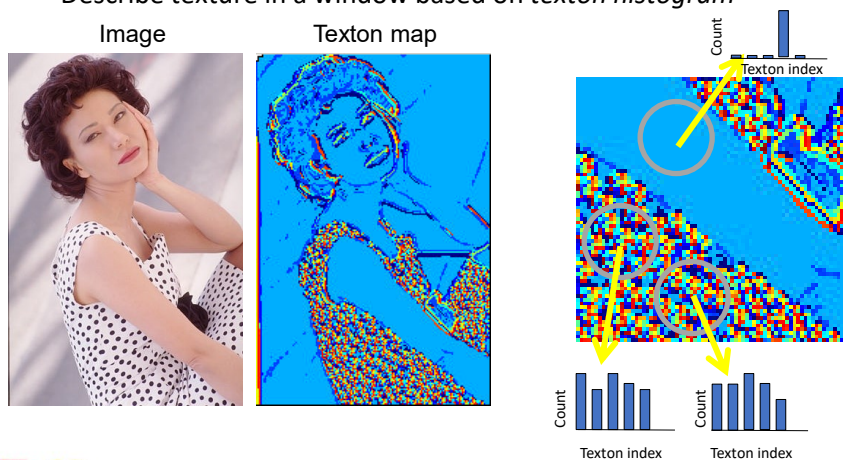
Feature space: filter bank responses (e.g., 24-d)

Slide credit: Kristen Grauman
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Slide credit: Kristen Grauman

Segmentation with texture features

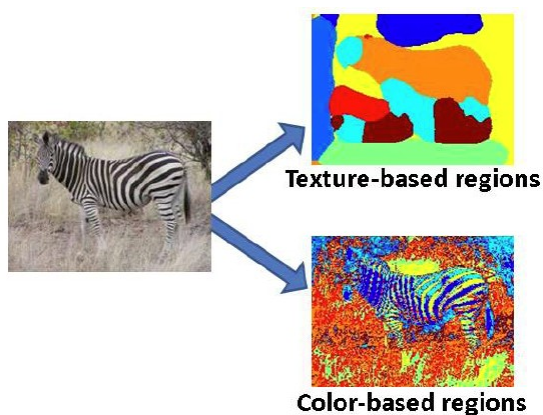
- Find “textons” by **clustering** vectors of filter bank outputs
- Describe texture in a window based on *texton histogram*



Malik, Belongie, Leung and Shi. IJCV 2001.
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Adapted from Lana Lazebnik

Image segmentation example



Slide credit: Kristen Grauman

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Region-based segmentation

- Finding region based on the criterion of **homogeneity** and **connectivity** of pixels (region)
 - Each region is homogeneous (i.e., uniform in terms of the pixel attributes such as intensity, color, range, or texture, etc.)
 - and connected
- **Algorithms:**
 - Region growing
 - Split and merge algorithm
 - Hierarchical clustering
 - ...

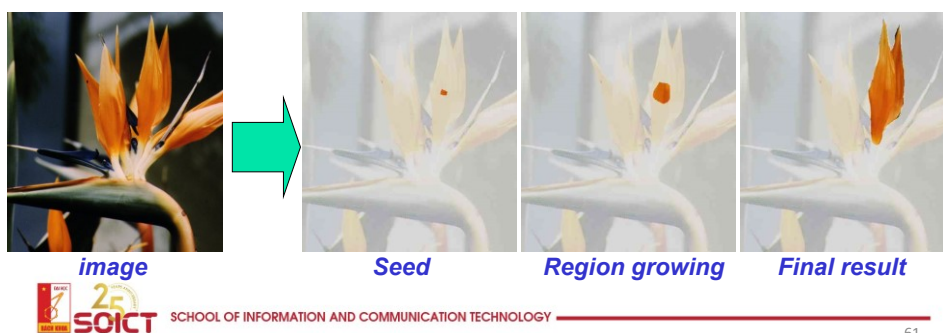


Region-based segmentation

- Region-based approaches provide :
 - All pixels must be assigned to regions
 - Each pixel must belong to a **single region** only
 - Each region must be **uniform**
 - Any **merged pair of adjacent regions** must **be non-uniform**
 - Each region must be **a connected set of pixels**
- Region-based approaches:
 - Different methods
 - Common point: **homogeneity criteria**

Region growing

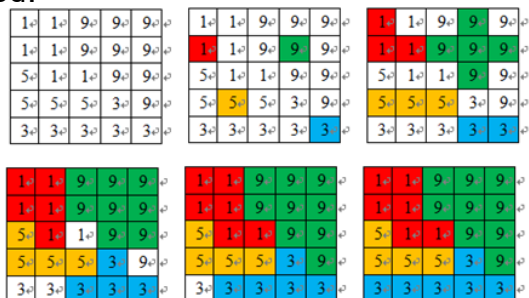
- Start from a point (seed) and add neighbor pixels following a **given criteria**
- The seeds can be manually or automatically chosen
 - automatic seeds in very homogeneous zones for example



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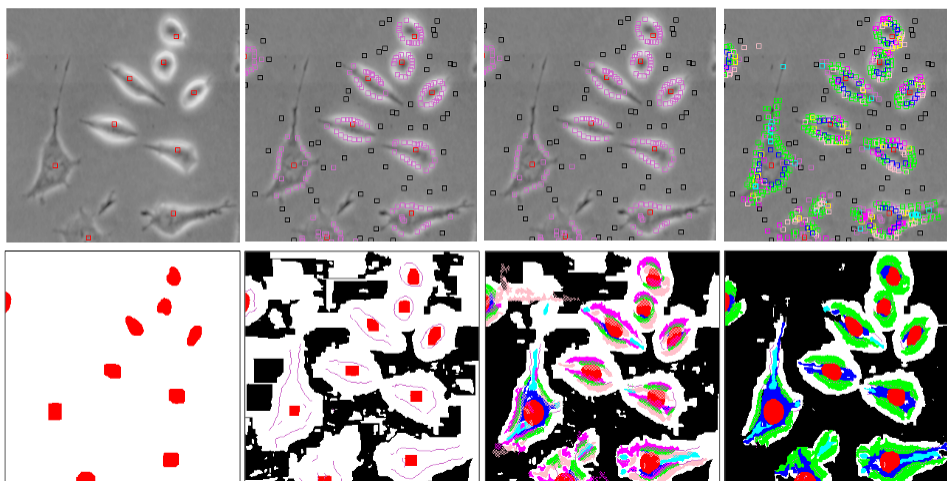
Region growing algorithm

- Algorithm:
 - Choose K random pixels in K regions
 - Use 4(8)-connected and threshold to determine
 - Repeat a and b until almost points are K classified.
- Example illustrated:



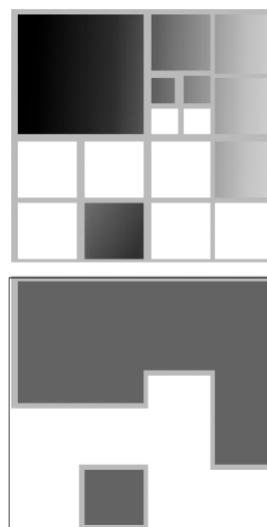
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Region growing with multi-seeds

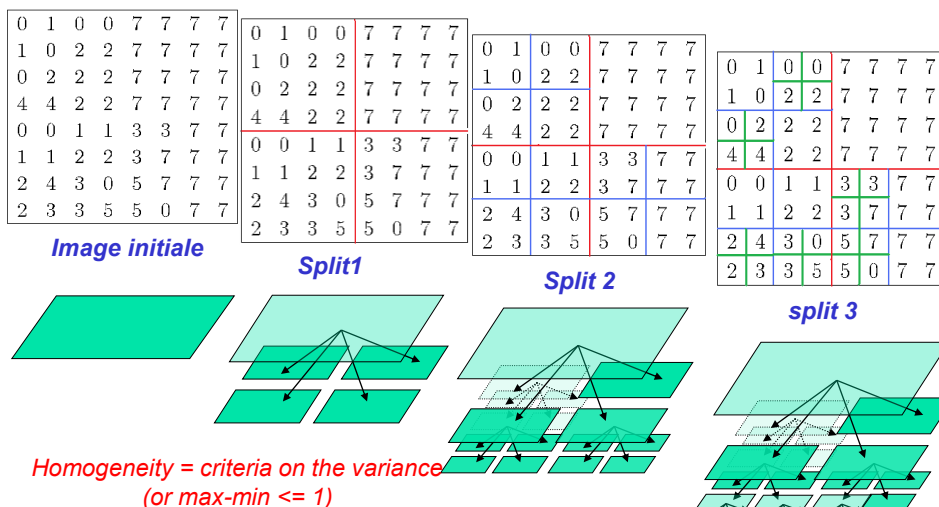


Split-and-merge

- Split (step 1)
 - Recursively split all non-homogeneous regions following a given criteria
 - variance, max-min, ...
 - Dividing one region gives 4 subregions
 - Subregion attributes are re-computed
- Merge (step 2)
 - Group all homogeneous adjacent regions following a given criteria



Split-and-merge: split

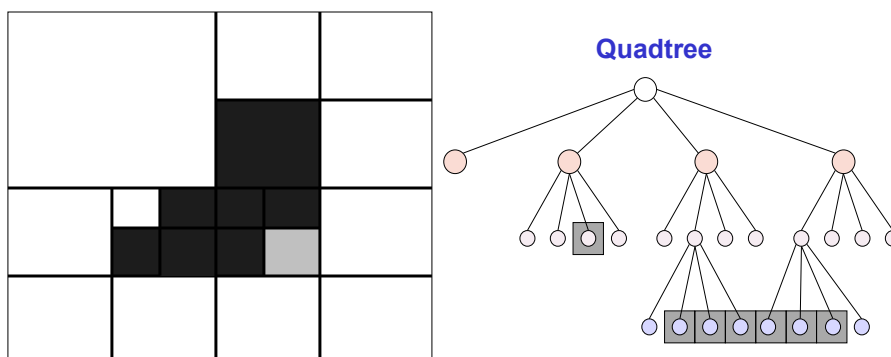


Source : Jean-Christophe Baillie. Cours de segmentation. ENSTA ParisTech (France).

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Split-and-merge: merge

Phase 1: Create homogeneous zones (split)
Phase 2: Group homogeneous zone (merge)



Connect homogeneous adjacent regions



Source : Jean-Christophe Baillie. Cours de segmentation. ENSTA ParisTech (France)

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Split-and-merge



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Edge-based segmentation

- Finding region based on edges

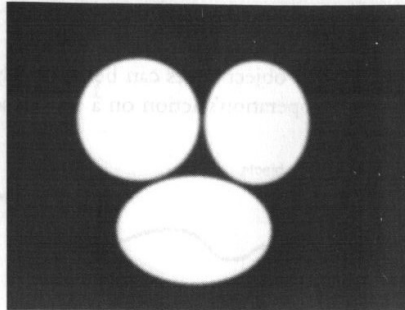


- **Algorithms:**
 - Basic Edge Detection
 - The Marr-Hildreth edge detector (LoG)
 - Short response Hilbert transform (SRHLT)
 - Watersheds

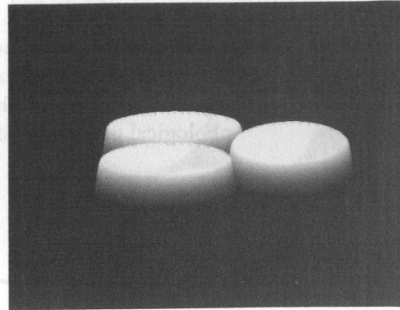
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Watershed segmentation

- We consider the image as a 3D shape using the gray level as the third dimension



2D image



Visualization in 3D



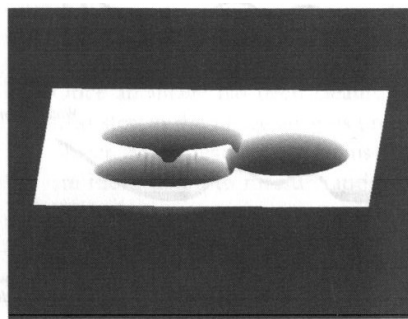
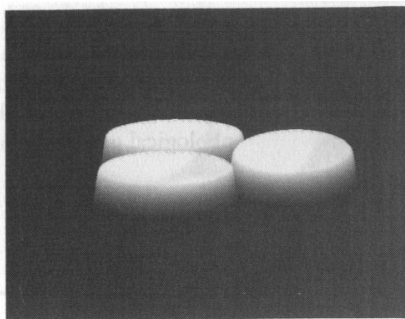
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Source : <http://www.gpa.etsmtl.ca/cours/gpa669/>

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Watershed segmentation

- After we reverse (upside down) the values to create « holes » in the shape

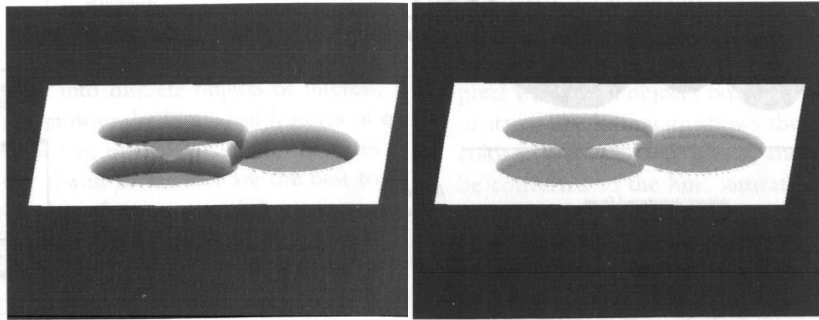


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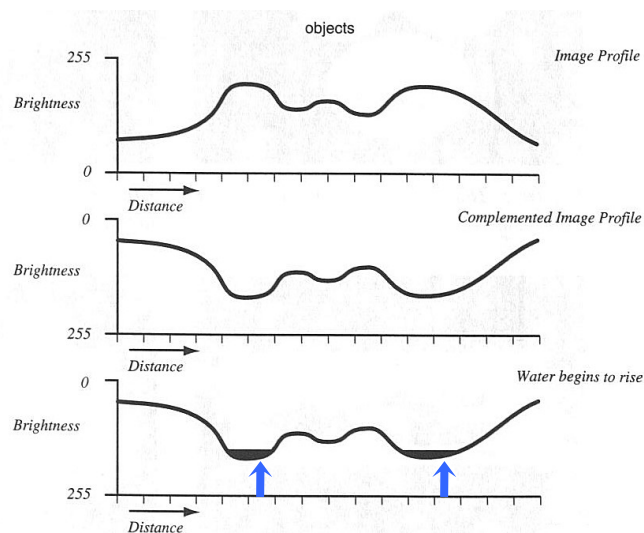
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Watershed segmentation

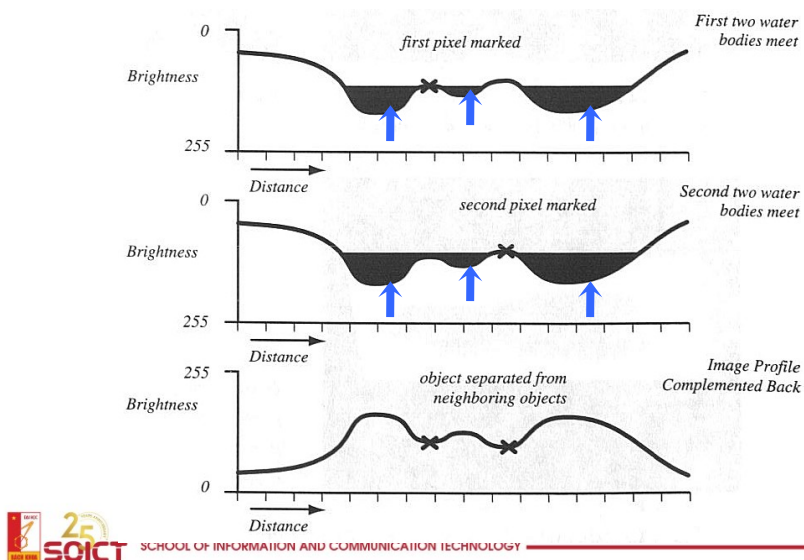
- Next we fill in the holes with water



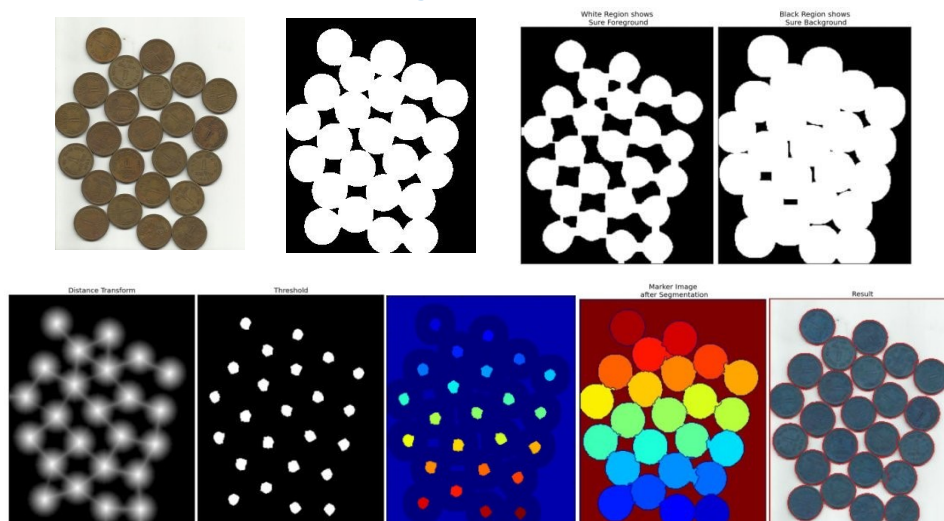
Watershed segmentation



Watershed segmentation



Watershed segmentation



Segmentation – advices

- Image segmentation
 - No method works for all images
 - No miracle, no warranty!
- One of the main problem is to define the **goal of segmentation**:
 - What exactly are we looking for in the image?
 - Global regions or small details?
 - Presence or not of persons details in the face?
- It is good to think in advance **what we will do with** this segmentation results
 - This helps to define the level of precision needed

Segmentation – advices

- Image Pre-processing:
 - **good selection** of sensors and energy source, and controled image acquisition conditions help to make segmentation easier and more efficient
- For some applications, we realize today that we can **avoid to segment** the image. It is often better like this.

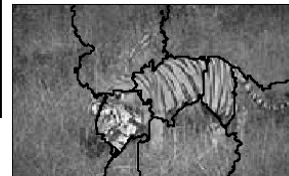
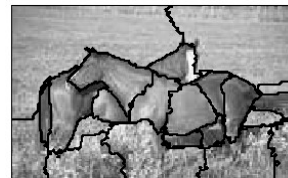
Limits of segmentation

Image segmentation alone cannot find all image objects as we can interpret them



Segmentation vs. grouping

- Term 'segmentation' :
 - less used
 - segmentation, which let think about an exact image splitting into regions
- 'Pixel grouping'
 - which refers only to a notion of similarity between pixels without relation on the content of regions.



Source : [Malik 2001].

Motion segmentation



Input sequence

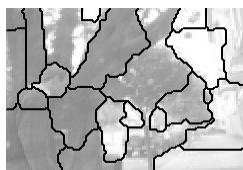


Image Segmentation



Motion Segmentation



Input sequence



Image Segmentation



Motion Segmentation

A.Barbu, S.C. Zhu. Generalizing Swendsen-Wang to sampling arbitrary posterior probabilities, *IEEE Trans. PAMI*, August 2005.

Credit: Kristen Grauman, UT Austin



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**Thank you for
your attention!**

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